Hariswar Thoguluva Prakash

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Summary

Data Engineer and Integrated Masters in Data Science candidate with proven industry internship experience engineering predictive analytics pipelines and enterprise data solutions. Experienced in langGraph agent orchestration, asynchronous OCR processing, database management across SQL/NoSQL systems,demand forecasting fine-tuning, automated inventory auditing engines, data structures, and statistical modeling, with additional hands-on experience in computer vision, deep learning, and Generative AI.

Education

Thiagarajar College Of Engineering — Madurai, TamilNadu
Integrated Masters in Data Science; CGPA: 8.24/10 — Expected May 2027

Adhyapana School — Madurai, TamilNadu
HSC; Percentage: 83.4 — April 2022

Adhyapana School — Madurai, TamilNadu
SSC; Percentage: 92.3 — April 2020

Skills

Programming Languages: C, Java, Python, MySQL
Computer Science & Systems: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks
Backend & Databases: MySQL, PostgreSQL, MongoDB, Neo4j,Rest API
Frontend Engineering: HTML, CSS
Tools: Git, Github, Tableau, PowerBI
AI/ML: Machine Learning, Generative AI, Large Language Models (LLMs), RAG, PyTorch, TensorFlow,Polars, Scikit-learn
Computer Vision: OpenCV, YOLO, OCR

Experience

ConverSight.ai — June 2025 – October 2025
Data Research Engineer Intern — *Coimbatore, TamilNadu*

- Engineered and fine-tuned predictive time-series models (SARIMAX, Prophet, XGBoost) in Python at Conversight.ai, integrating regional holidays and festival calendars as exogenous variables to improve demand forecasting accuracy by 24% and reduce Mean Absolute Percentage Error (MAPE) by 18%.
- Architected automated feature engineering and selection pipelines using Polars, Scikit-learn, and PostgreSQL, streamlining high-volume historical data ingestion and reducing execution latency by 35%.
- Designed and deployed interactive executive analytical dashboards in Conversight platform for client demo sessions and academic review, converting complex inventory metrics into visual intelligence and reducing stock anomaly detection time by 40%.
- Benchmarked predictive algorithms against enterprise inventory datasets, optimizing hyperparameter tuning to eliminate stockout risks and prevent procurement inventory leakage

Adhish Technologies Private Limited — December 2024 – January 2025
Engineering Intern — *Madurai, TamilNadu*

- Evaluated and benchmarked deep learning and open-source OCR architectures (TrOCR, EasyOCR, Pytesseract) across complex industrial parts datasets, identifying optimal computer vision models to increase character extraction accuracy by 22% under low-resolution and noisy lighting conditions.
- Engineered an automated computer vision preprocessing and text-extraction pipeline in Python, optimizing image binarization, noise filtering, and bounding-box segmentation to reduce label recognition errors and accelerate downstream data parsing.

Projects

Autonomous AI Incident Responder Platform

- Architected an event-driven incident investigation platform using Python, FastAPI, and PostgreSQL to ingest real-time webhooks and orchestrate parallel evidence gathering across OpenTelemetry traces and GitHub code diffs.
- Engineered a self-correcting LangGraph agentic workflow featuring reflection loops and context-pruning strategies, eliminating LLM hallucinations during root-cause analysis while reducing token operational costs by 80%.

AI powered Reconciliation Platform For Textile Manufacturing

- Architected an asynchronous OCR pipeline using Python, Celery, Redis, and PostgreSQL to parse weighbridge receipts and invoices, accelerating document processing speed and eliminating manual data entry errors.
- Engineered a dynamic, moisture-adjusted auditing engine to automatically detect raw material inventory shortfalls and issue vendor debit notes, eliminating procurement revenue leakage.

Certifications

- **Artificial Intelligence and Machine Learning** – IIT Madras
- **Data Mining Methods & Mathematics for Linear Algebra** – Coursera

LEADERSHIP AND ACHIEVEMENTS

- **Open Source Contribution – OpenSRE:** Implemented configurable Sentry logging integration, added unit tests for error reporting, and contributed to CLI documentation and reference updates.
- **1st Place – Mathematical & Statistical Contest & C Debugging Contest:** Secured top rank in competitive technical challenges for statistical analysis, algorithmic logic, and rapid troubleshooting.
- **3rd Place – Machine Learning Contest:** Recognized for exceptional performance in statistical analysis, and predictive modeling.
- **Technical Lead – IDEA Magazine:** Directed content curation and documentation for the annual publication, publishing student research, academic milestones,achievements and department progress.